

A Single-Exponential Exponent for Arithmetic Progressions

AI-assisted research draft^{*}

8 October 2026

Abstract

Using the analytic inputs of *Quasipolynomial Bounds for Arithmetic Progressions* (OpenAI/math, family 159), we obtain an admissible density exponent $\varepsilon_k = \exp(-k^C)$ for an absolute constant C . Two changes give the single-exponential dependence. An ordered return factorization keeps historical substitutions out of the next active orbit, so the geometric degree is $O(k^2)$. An inverse-theorem induction retains the supplied derivative witnesses, so its logarithmic-budget exponent is single exponential in the order. The original rank bookkeeping and a descending width recurrence then suffice. Only the single-exponential dependence on k is tracked.

1 The result and its inputs

Let $r_k(N)$ be the largest size of a subset of $[N]$ containing no nonconstant k -term arithmetic progression. Write $\mathcal{M}$ for the September 23, 2026 manuscript [1].

Theorem 1.1. *Assume the analytic inputs of $\mathcal{M}$ listed below. There is an absolute constant $C > 0$ such that, for every $k \geq 3$, there are constants $K_k, c_k > 0$ for which*

$$r_k(N) \leq K_k N \exp[-c_k(\log N)^{\varepsilon_k}], \quad \varepsilon_k = \exp(-k^C), \quad N \geq 3.$$

The inputs are the bottom absolute-patch theorem (Lemma 2.2), the sampler and rank-transfer estimates (Sections 3–4 and Appendices F–H), the field and active-return framework (Definition 8.1 and Theorem 8.2), the exact macro extraction (Section 9), and the compression and descending numerical schedule (Section 10). The inverse-theorem input is the argument of Leng–Sah–Sawhney [2], with its quantitative induction modified in Section 4. We prove the additional algebraic and quantitative assertions required for these modifications. The conclusion retains this dependence on the analytic inputs of $\mathcal{M}$.

Throughout, constants depending only on k , or on a fixed theorem order, may be absorbed into multiplicative constants. Powers of growing budgets are retained. Thus a bound $K_D x^{\exp(D^{O(1)})}$ is acceptable, whereas an uncontrolled iteration of such a power is not. Every $O(1)$ in an exponent below is absolute.

For a starting density $\alpha > 0$, put

$$s = k - 2, \quad D = 256k^2, \quad p = 2 + \log(1/\alpha).$$

The density increment takes $T \ll_k p$ rounds. At weight h , the determining data have integer coordinates b_h , a rational value space W_h , and targets

$$C_h(u, b_{<h}) \in W_h, \quad \deg_{\text{wt}} C_h \leq h.$$

The spatial variable has weight one and b_h has weight h . The strict and enlarged determining cutoffs force a unique integer tuple. If w_h is the width, write $Q_h = \log(2/w_h)$. The essential order condition is that a width loss at h may depend on numerical data already fixed at weights $> h$, but not on incoming Q_i with $i \leq h$. We call such a loss *warm at h* . Data using the current

^{*}AI assistance: OpenAI ChatGPT was used for mathematical exploration and drafting; OpenAI Codex was used for editing and compilation.

Q_h are allowed in a preliminary bound at h ; when retained as history, they are warm at every later weight $j < h$.

The proof has three quantitative outputs:

$$\text{geometric degree } D = O(k^2), \quad \text{local budget degree } B \leq \exp(D^{O(1)}), \quad \varepsilon_k^{-1} \leq \exp(D^{O(1)}).$$

We keep the original rank-cut bookkeeping; no improvement to that bookkeeping is needed.

2 A persistent record with quadratic degree

A completed active return has two kinds of controlled factors: a slow factor in normalized coordinates and a rational factor in integer coordinates. Keeping these factors on their original sides allows the middle orbit to retain its original degree.

For a filtered Lie group, call a polynomial *jointly filtered* when each monomial of weighted degree a in its logarithm has coefficient in the a -th Lie layer. Constants lie in the first layer. Integer and normalized labels have weight one. A *marking* is the fixed projection $\pi : G \rightarrow Y$ recording the determining equations.

Lemma 2.1 (Ordered active return). *In the setting of Theorem 8.2 of $\mathcal{M}$, suppose the input orbit is jointly filtered in all actual parameters and its marking is a fixed filtered polynomial $P_Y(X)$ of the site variables. The group and marking may themselves be finite charged families in older dummy variables, which are coefficients rather than actual arguments. The return can be chosen in the form below, with each of E, C, R jointly filtered in the original filtration:*

$$g^\sharp(X, \theta, \tilde{t}, z) = E(X, \theta, z) C(X, z) R(X, \tilde{t}, z), \quad \pi E = \pi R = 1, \quad \pi C = P_Y(X). \quad (1)$$

Here X collects the site variables, z the short labels, and $\theta, \tilde{t}$ the normalized and integer long labels. The factor C has no long-label argument; E has controlled coefficients in normalized site, residual, and label coordinates; and R has controlled common denominator in raw integer coordinates. No coefficient bound is required for C . The original score slice, both expected-payoff representative identities, and the field estimates are retained.

The additional induction proving the lemma is given in Section 3. We first show how its format closes the macro argument.

Proposition 2.2 (Persistent macro record). *Starting from an absolute patch of degree s and rank d_0 , the macro argument can retain the original d_0 absolute slots and a middle orbit of joint degree at most s . Its final physical formula has absolute-weight ceiling at most $s(s+1)$. The representation, field, score, and extraction contracts of $\mathcal{M}$ are preserved, with warm costs at each later layer. The padded ceiling $D = 256k^2$ therefore suffices.*

Proof. The retained data. Store the return as $E_{\text{hist}} C R_{\text{hist}}$, and use only C as the next active orbit. Historical integer short labels remain formal coefficient variables, with *one total-charge cap* s over all their blocks. Raw and normalized copies of a short label always use the same variable and its fixed normalizing divisor. Historical long labels remain in the outer factors. A deep coefficient lattice evaluates into the pointwise lattice at every integer short label; arbitrary real labels have no such property. Before the next positive-filtration recursion, normalize any degree-zero family constant as in the proof of Lemma 2.1.

A newly promoted determining coordinate retains its original weight in C . In outer-factor coefficients use its canonical site value: the left factor uses the bounded normalized residual frame, and the right factor uses raw integer coordinates. Exact marking identifies these values at every tested point. Fix the integral determining reindexings before recording history, and keep

their exact triangular inverses. They preserve the original determining degrees. Later congruence restrictions have their creation-layer costs, hence warm costs at every lower layer.

Ordered representation. Normalize the middle factor only. For an integral triangular reindexing T of the absolute tuple, the correct expression is

$$E_{\text{hist}}(CT)(T^{-1}R_{\text{hist}}T). \quad (2)$$

This is the old physical map followed by T , which permutes the complete integer tuple sum. Conjugation by the integral map preserves controlled rational denominators; its numerator need not be bounded. Coefficientwise reduction gives $CT = Sg$, where S is the bounded full-weight map. It is independent of the historical variables: their positive degree excludes them from a full-weight slot term. If $g = E_{\text{new}}C_{\text{new}}R_{\text{new}}$, collect the result as

$$(E_{\text{hist}} \text{Ad}_S E_{\text{new}})(SC_{\text{new}})(R_{\text{new}}R'_{\text{hist}}), \quad R'_{\text{hist}} = T^{-1}R_{\text{hist}}T. \quad (3)$$

Only bounded maps conjugate slow factors. Rational right factors stay on the right. The triangular subgroup in which determining transformations have no absolute arguments is bracket closed; forgetting the absolute slots is strong, since a determining shear lifts while fixing them. Thus this normalization preserves the exact determining projection.

The quotient fields. At site u , define the field on the coefficient-family quotient by the original cutoff sum evaluated on

$$E_{\text{site}}(u) S \text{ev}_{w(u)}(Y) R_{\text{site}}(u). \quad (4)$$

Canonical site substitution occurs only in the external coefficients and in $\text{ev}_{w(u)}$. All original determining and absolute integer variables are still summed. On an exactly marked orbit, any nonzero determining cutoff forces the unique canonical tuple. Hence both payoffs agree exactly with the originals, including their zero cases.

A common deep coefficient lattice, further deepened for the rational right factors, makes (4) well defined. On bounded group representatives, the left map and S are bounded; the right map can be reduced to a bounded right-coset representative determined by its denominator and coefficient residues. The triangular inverse in the representation argument then bounds all contributing integer tuples and derivatives. Bounded-range integer evaluation has controlled Lipschitz norm. This gives the required Lipschitz bound, independently of the unrestricted coefficients of C . Triangular uniqueness preserves the cap one.

The two field meshes. After extraction, product-column labels are frozen. The only historical short labels that remain variable are integral affine maps from extra inactive rows. Those rows annihilate the prepared value space. The chart estimate has the form

$$\|z_i\| \leq \exp(\text{poly}(d))K_i, \quad K_i \leq k_i/v,$$

where v is the chart width. For a short axis, its scale k_i , the integral offset, and its values throughout the fixed padded chart have preliminary bounds. The factor v^{-1} is included in that bound. At every later layer all these costs are warm. The fixed determining polynomials therefore turn the finite row-and-offset menu into a finite menu of entire functions $u \mapsto w(u)$.

Historical real labels are affine expressions in the current normalized spatial coordinates and the fixed higher normalized residuals. They are bounded on the same padded chart, even when they leave a formerly selected unit interval. Retain the original smooth cutoffs and score slice; no new moving sharp indicator is inserted. Outside this fixed padded chart use the zero extension from the representation construction of $\mathcal{M}$.

At one site, mesh bounded left values, full-weight coefficients, bounded integer evaluations, right cosets, and inherited cutoff parameters. For a mesh uniform over all sites, mesh the bounded left *coefficient vectors*, the finite whole-function short menu, rational coefficient residues, and the

inherited field parameters. The canonical site functions are fixed throughout. These give the distinct point-value and full-field meshes required by Definition 8.1 of $\mathcal{M}$.

First pigeonhole the finite tuple of actual historical denominators and cover moduli, then take their product and a fixed deepening. Its logarithm is a sum of warm budgets. Taking the least common multiple of all integers below an upper bound is unnecessary. After product-column freezing, historical raw labels are affine in raw spatial and determining integers and the remaining extra inactive labels. Later integer sections may have arbitrarily large coefficients; they change only the rational numerators, which are recorded modulo the common deep grid. The same divisibility gives the long-label periods. Selection pays a warm loss in the fraction of available productive paths; every retained path keeps its original full score and slice.

Extraction and the final degree. For a macro comparison, reassemble the physical formula. On its scored locus, replace canonical integers in external coefficients by their corresponding integer summation variables. Exact marking and the original cutoffs identify the expressions. Current auxiliary labels now enter only through this polynomial formula, and both expected representative identities apply before the stochastic constants are frozen. Section 9 of $\mathcal{M}$ uses these identities, the original slice and masks, cap one, exact determining and inactive equations, and a degree ceiling. It does not require bounds on coefficients or spatial derivatives of the returned physical polynomial. Its extraction and localization therefore apply with the original payoffs.

Every recorded historical-label substitution has final spatial/determining weighted degree at most s . For a monomial of original absolute-input weight h' and other charge d , the joint invariant gives $h' + d \leq h$, where $h \leq s$ is its absolute destination weight. Give each absolute slot the temporary weight $(s + 1)h$. Simultaneous substitution gives

$$(s + 1)h' + sd \leq (s + 1)(h' + d) \leq (s + 1)h. \quad (5)$$

This reweights absolute inputs and destinations together. Consequently the physical ceiling is $s(s + 1)$.

These temporary weights are used for extraction and final compression. The next middle orbit still uses the original weights, with determining slots and historical integer variables formal; its outer factors again enter the fields. This closes the induction without repeated degree inflation. No family or partner coordinate becomes an absolute slot, so the root rank is still d_0 , independent of old numerical widths.

Finally, m historical variables of total charge at most s have at most

$$\binom{m + s}{s} \leq (m + 1)^s$$

monomials per scalar coordinate. There are at most s macro layers and s internal degree reductions per return. Rational systems, coefficient arrays, and finite meshes are handled simultaneously. The added operations have polynomial sequential depth in s ; their quantitative bounds are included in Section 4. $\square$

3 The charged active return

We prove Lemma 2.1 by supplementing the degree induction of Theorem 8.2 of $\mathcal{M}$. The point is to retain total degree when short variables become coefficient variables, while each recursive child still has strictly smaller positive degree.

Proof of Lemma 2.1. Charged families and positive normalization. Record rational coefficient spaces $A_{d,\ell}$ with

$$[A_{d,\ell}, A_{e,m}] \subset A_{d+e,\ell+m}, \quad A_{0,\ell} = A_{1,\ell}, \quad A_{r+1,\ell} = 0,$$

and one finite total-charge cutoff in ℓ . Initially $A_{d,\ell} = \mathfrak{g}_{\max(d,\ell)}$, with $\mathfrak{g}_0 = \mathfrak{g}_1$. Promoting a short block to a dummy w replaces the array by

$$A'_{d,\ell} = \left\{ \sum_{\beta} w^{\beta} a_{\beta} : a_{\beta} \in A_{d,\ell+|\beta|} \right\}. \quad (6)$$

The positive layers are $\exp A'_{i,i}$, and the degree-zero group is $\exp A'_{0,0}$. Quotients use images of these spaces, and refinements use the actual intersections before taking images. Every raw and normalized occurrence of a promoted label is changed simultaneously, including those in marks and ancestor factors.

Perform the common-site marking correction before fast-algebra selection. A coefficientwise ordinary filtered section commutes with evaluation at the actual short label, so the inserted correction evaluates to the identity. Normalize the fixed marked constant by a bounded left factor and a rational right factor. The normalized mark takes values in the positive marked group. Let $c = g(0, 0; w)$ be the constant family element after this marked normalization. The remaining constant coset modulo the positive group can now be removed in the kernel: choose a positive-layer lift c_1 of its marked image, and split the difference $n_0 = cc_1^{-1} = e_N \gamma_N$ into a bounded kernel element and a kernel lattice element. Replace the family orbit g by $e_N^{-1} g \gamma_N^{-1}$, retaining both factors for restoration. This lies in the positive group because its constant coset does. These are family-group operations, so all charge bounds are retained. A deep source lattice with the prescribed image handles the marked rational factor; its numerator need not be bounded.

Refiltration with the old degree retained. The degree-zero action is made compatible with the refiltration by adjoining to the detected fast algebra the rational graded ideal

$$S_i = \text{image} \left(A'_{i+1,i} \longrightarrow A'_{i,i} / A'_{i+1,i+1} \right).$$

The bracket rule makes degree-zero conjugation trivial modulo S_i , and $S_r = 0$. Thus saturation preserves the detected top and its injective marked projection, while the resulting refiltration is normalized by the degree-zero group.

Write G_i for the old positive layers and H_i for the refiltration from Lemma 7.6 of $\mathcal{M}$. The prescribed full factors preserve old joint degree: a grade- i symbol correction is lifted into G_i ; the off-space substitution preserves weighted degree; and the constant adjustment sees only old grade one, the higher short degrees already lying in $G_2 \subset H_1$. A pure-long grade- i symbol is independent of the short variables, since every coefficient of positive short degree lies in old G_{i+1} ; its kernel lift therefore preserves total degree. Filtered multiplication and conjugation preserve these bounds. The refiltered input therefore satisfies

$$[v^{\alpha} z^{\beta}] \log h \in \mathfrak{h}_{|\alpha|} \cap \mathfrak{g}_{|\alpha|+|\beta|}, \quad \mathfrak{h}_0 = \mathfrak{h}_1. \quad (7)$$

Here v denotes the long variables and z the current short variables.

Since $G_{i+1} \subset H_i \subset G_i$, the layers

$$H_1 \supset G_2 \supset H_2 \supset G_3 \supset \cdots$$

form one rational flag, with its strong image in the marked group. A rational section adapted to this flag preserves the required intersections. It gives a fixed exact lift $p(X, z)$ of the refiltered mark with

$$[X^{\gamma} z^{\beta}] \log p \in \mathfrak{h}_d \cap \mathfrak{g}_{d+|\beta|}, \quad d = \text{wt } \gamma.$$

Only this interleaved local flag is used; historical charge is already encoded in the coefficient-family group.

One lower-degree child. Let $q : H \rightarrow Q = H/H_r$, and let N be the actual kernel of the marking on H . Top injectivity identifies N with the kernel of the quotient marking via $q|_N$. Promote the current short block to a new dummy w and form

$$\mathcal{Q}_i = \left\{ \sum_{\beta} w^{\beta} a_{\beta} : a_{\beta} \in q(\mathfrak{h}_i \cap \mathfrak{g}_{i+|\beta|}) \right\},$$

$$\mathcal{N}_i = \left\{ \sum_{\beta} w^{\beta} n_{\beta} : n_{\beta} \in q(\mathfrak{n} \cap \mathfrak{h}_i \cap \mathfrak{g}_{i+|\beta|}) \right\}.$$

At index zero use $\mathfrak{h}_0 = \mathfrak{h}_1$ and $\mathfrak{g}_0 = \mathfrak{g}_1$. These are images of the displayed intersections. The bracket relation $[\mathcal{Q}_i, \mathcal{N}_j] \subset \mathcal{N}_{i+j}$ defines the split rational family $\mathcal{N} \rtimes \mathcal{Q}$, marked by its second coordinate. Its canonical section is $a \mapsto (1, a)$. The map $(n, a) \mapsto (na, a)$ embeds it into two copies of the quotient family, so it is a rational nilpotent group. Both degree- r spaces vanish, so its positive group has degree at most $r - 1$.

The local pair

$$(qh(qp)^{-1}, qp)$$

has the required kernel coefficients by (7). Correct its marking to the common site with the canonical section, and apply the degree-zero normalization above. The resulting input is jointly filtered in all actual variables: substituting a weight- d site monomial can produce actual degree $a \leq d$, and a dummy degree b is charged in the old layer $\mathfrak{g}_{d+b} \subset \mathfrak{g}_{a+b}$. Apply the induction to this single child, leaving w unevaluated.

Restore the child's preliminary degree-zero factors, so its exact second coordinate is $qp(X, w)$. Multiplying by the inverse of its canonical marked section isolates the kernel family. Lift this family through $(q|_N)^{-1}$, multiply by $p(X, w)$, and evaluate $w = z$. Its degree- a , dummy-degree- b coefficients lift into $\mathfrak{n} \cap \mathfrak{h}_a \cap \mathfrak{g}_{a+b}$. Thus reconstruction restores the original joint degree exactly, including synchronized normalized copies. A common deep coefficient lattice evaluates inside the pointwise lattice at every integer z . Fields pull back through the homomorphism $(n, a) \mapsto na$; pullback preserves supremum meshes, and bounded short evaluation has controlled Lipschitz cost. No metric bound on the particular marked lift p is used.

Ordered factors and the original score. Keep the child return in the full split marked group. The joint-fiber reconstruction T is equivariant for actual kernel factors:

$$T((e, 1)c(r, 1)) = \tilde{e}T(c)\tilde{r}.$$

Indeed the two sides have the same quotient and exact mark, and the joint embedding is injective. The lifts $\tilde{e}, \tilde{r}$ are rational and filtered. Collect them through the prescribed bounded left and rational right factors of $\mathcal{M}$. For the local notation $K_E \rho(E_F)$ and $\rho(R_F)K_R$, this gives

$$E = K_E \text{Ad}_{\rho(E_F)}(\tilde{e}), \quad C = \rho(E_F)T(c)\rho(R_F), \quad R = \text{Ad}_{\rho(R_F)^{-1}}(\tilde{r})K_R.$$

The left conjugation has bounded normalized coefficients, and the right one has controlled rational denominator. The middle has no long-label argument.

If L is the marking restorer of $\mathcal{M}$, use $E(LC)R$. It has the required marking everywhere and agrees with $LECR$ on the full exact-inactive forecasting locus, where $L = 1$. That is the locus of both original payoffs and their representative identities. At an earlier central quotient, lift the outer factors separately in the actual kernel and lift the middle separately. Any difference from the original selected lift is removed by the independent central Haar average. Restore the initial degree-zero factors and project away partners. The marking and joint degree are exact, and the expected payoffs, original slice, and additive analytic errors are those of $\mathcal{M}$.

At degree zero the scalar-comparison base case applies. Every positive step has one strictly smaller-degree child. This proves the lemma. All additional operations are simultaneous rational linear algebra, bounded-degree coefficient constructions, and one split product per step. $\square$

4 Single-exponential numerical costs

There are two separate numerical tasks. The external inverse theorem must have a single-exponential *budget exponent*; the operations inside $\mathcal{M}$ must preserve that class.

4.1 The external inverse exponent

Let I_s be an exponent such that the degree- s inverse theorem produces a correlating nilsequence with

$$d + \log M + \log(2 + \|F\|_{\text{Lip}}) + \log(1/\text{corr}) \leq K_s(2 + \log(1/\delta))^{I_s}. \quad (8)$$

Here d is the nilmanifold dimension, M its complexity, and δ the Gowers-norm threshold.

Proposition 4.1. *The inverse-theorem argument can be run with*

$$I_s \leq \exp(s^{O(1)}).$$

Proof. Dimension and logarithmic height. Keep d separate from the logarithmic numerical budget Λ , which includes rational heights, reciprocal errors, Lipschitz norms, and the logarithm of the nilcharacter's output dimension. Basic rational and coordinate operations in order q have the form

$$d_{\text{out}} \leq K_q d, \quad \Lambda_{\text{out}} \leq K_q(2 + d)^b \Lambda_{\text{in}}. \quad (9)$$

Common denominators are cleared jointly; maximal minors give bases, kernels, and sections with logarithmic height polynomial in d and linear in Λ . For bounded-step BCH, all bracket denominators divide a fixed power of the common structure-constant denominator. Fourier term counts and bounded-step products likewise have logarithmic cost polynomial in d and linear in Λ .

In Leng's degree/step induction [3], each recursive group is a subgroup or a square-group quotient of dimension at most $2d$. The next recursive call uses the new logarithmic budget but a dimension comparable to the original d . Thus its dimension exponents satisfy

$$b_{a,b} \leq b_{a-1,b} + b_{a,b-1} + q^{O(1)}, \quad b_{a,1}, b_{1,b} \leq q^{O(1)}.$$

Pascal induction gives $b_{q,q} \leq \exp(O(q))$. The later step reductions of [2, Theorem 5.4] stay in subgroups or quotients and add such exponents. The simultaneous denominator bounds in the rational calculus handle the dimension loops without repeated height substitution.

Take $q = O(s^2)$ for the degrees and rank filtrations appearing in [2]. Its universal groups have dimension at most $K_q m^q$ on m generators. Polarization and the constructions for powers and roots use an order-dependent number of copies, and the latter use subgroups of $G \times G$; they do not create a dimension power equal to the integer root order. Chart counts and nilcharacter output coordinates are charged logarithmically. The finite helper chains, degree-rank passages, and converse-inverse reductions have polynomial sequential depth in s . Consequently the non-inverse part of an order- s construction has budget degree

$$F_s \leq \exp(s^{O(1)}) \quad (10)$$

in $2 + d + \Lambda$.

The additive-combinatorial primitive and the fixed U^3 base case have polynomial logarithmic budgets: the Bogolyubov–Ruzsa and progression-to-Bohr steps are [7, 6], the quadratic construction is [4, Appendix A], and its nilsequence conversion is [5]. In the last conversion, the group has dimension $O(m^2)$, the radius loss has logarithm $O(m \log m)$, and the defining cutoff sum has at most 3^m terms on a fundamental cube. Thus its logarithmic Lipschitz cost is polynomial as well. All these fixed-order costs can be included in F_s .

Keeping derivative witnesses. We need a stronger induction than the ordinary inverse statement. Let J_s assert that if a 1-bounded function $f : [N] \rightarrow \mathbb{C}$ has, for some $H \subset [N]$ with $|H| \geq \rho N$ and $0 < \rho \leq 1$, supplied degree- $(s-1)$ nilsequences ν_h such that

$$\left| \mathbb{E}_{n \in [N]} \Delta_h f(n) \nu_h(n) \right| \geq \rho, \quad \Delta_h f(n) = f(n) \overline{f(n+h)},$$

then f has a degree- s correlating nilsequence with budget at most $K_s x^{J_s}$, where

$$x = 2 + d + \log M + \log(1/\rho).$$

We use the zero extension of f . The witnesses may have different nilmanifolds and functions, each of dimension at most d , complexity at most M , and Lipschitz norm at most M . They are normalized to be 1-bounded.

The elementary step is the following consequence of Cauchy–Schwarz. If

$$\mathbb{E}_{h \in [L]} \left| \mathbb{E}_{n \in [L]} F(n) G(n+h) \tau_h(n) \right| \geq \eta \quad (11)$$

where F is 1-bounded and zero off $[L]$, G is 1-bounded on $[2L]$, and each τ_h is a 1-bounded degree- u nilsequence with dimension at most d , complexity at most M , and Lipschitz norm at most M , then, when $L \gg \eta^{-2}$, at least $c\eta^2 L$ positive shifts a have supplied witnesses σ_a with

$$\left| \mathbb{E}_{n \in [L]} F(n) \overline{F(n+a)} \sigma_a(n) \right| \geq c\eta^2. \quad (12)$$

Here σ_a is the product of two shifted τ_h 's, one conjugated, so its degree is unchanged and its dimension at most doubles; its logarithmic complexity and Lipschitz costs remain controlled.

For completeness, choose unit scalars ω_h aligning (11) and set

$$U(m) = \sum_{h=1}^L F(m-h) \omega_h \tau_h(m-h).$$

Cauchy–Schwarz gives $\sum_m |U(m)|^2 \geq \eta^2 L^3 / 2$. The diagonal terms contribute at most L^2 . Pairing positive and negative differences in the remaining sum yields

$$\sum_{a=1}^{L-1} \max_{1 \leq b \leq L-a} \left| \frac{1}{L} \sum_n F(n) \overline{F(n+a)} \tau_{b+a}(n) \overline{\tau_b(n+a)} \right| \geq c' \eta^2 L.$$

Each summand is at most one, so a proportion $\gg \eta^2$ exceeds $c\eta^2$. Choose b separately for each a ; there is no loss for the number of available witnesses. Unbounded shifts enter the polynomial orbit, not the nilmanifold function. If $L \ll \eta^{-2}$, Parseval and $\mathbb{E}|F|^2 \geq \eta^2$ instead give a linear-phase correlation of size at least $\eta/\sqrt{L}$.

The non-inverse construction of [2, Lemma 6.2] accepts the supplied family directly: omit the paragraph that obtains the witnesses from the derivative Gowers norm, and retain its normalization, structure-constant pigeonhole, common-function selection, and vertical Fourier steps. These use only d, M, ρ . The subsequent correlation-structure induction (Lemma 6.3 through Theorem 6.4) and the integration construction in Section 12 use no further ordinary inverse theorem before its final Gowers/inverse steps. Their budget is $K_s x^{F_s}$.

Immediately before those final steps, the argument supplies (11), with

$$F(n) = f(sn + q_0) A(n), \quad \deg A \leq s, \quad \deg \tau_h \leq s-2.$$

The known factor A includes the nilsequence multiplier already present in that correlation. Normalize all factors within the existing logarithmic budget. Applying (12) gives exactly the supplied-family hypothesis for J_{s-1} , with input size at most $K_s x^{F_s}$. Its output Θ gives the final

test $A\Theta$, of degree at most s , against f on the selected residue class. The final residue-class Fourier expansion in [2] returns the correlation on $[N]$. In particular, the known factor A is retained throughout.

Degree one is elementary. The base J_2 follows from the fixed U^3 case after applying the converse bound to the supplied linear witnesses. If a size condition fails at any stage, $\log N \leq K_s x^{F_s}$; the available correlation gives a lower bound on $\mathbb{E}|f|^2$, and Fourier analysis supplies the same logarithmic-budget alternative.

Enlarge F_s within the class (10) to include initialization and final assembly. Sufficient recurrences are

$$J_s \leq F_s(J_{s-1} + 1), \quad I_s \leq F_s(J_s + 1)(I_{s-1} + 1). \quad (13)$$

The second recurrence first uses the ordinary degree- $(s-1)$ inverse theorem once to supply derivative witnesses, then uses J_s . Taking logarithms and summing, with $\log F_s \ll s^c$ for an absolute c , gives

$$\log(J_s + 1) \ll s^{c+1}, \quad \log(I_s + 1) \ll s^{c+2}.$$

This proves the proposition. $\square$

4.2 The numerical library

Proposition 4.2. *For the construction of Sections 2–3, each round introduces at most $K_k p^F$ fresh absolute coordinates, and the width increments, root logarithmic costs, and logarithmic side requirements have a common polynomial degree B , with*

$$1 \leq F \leq B, \quad 2 \leq B \leq \exp(D^{O(1)}). \quad (14)$$

The width increment at weight i uses only width logarithms Q_j with $j > i$.

Proof. Let $x \geq 2$ dominate the input dimensions and the logarithms of rational heights, function complexities, reciprocal errors, and scale requests. Simultaneous rational elimination, bounded-step BCH, coefficient arrays, Fourier truncation, and finite-dimensional meshes have degree at most

$$H_D \leq \exp(D^{O(1)})$$

in x . An external inverse call has degree at most $H_D(I+1)$, where $I = \max_{r \leq D+2} I_r$. Proposition 4.1 gives $\log(I+1) \leq D^{O(1)}$.

Two distinctions prevent artificial exponentiation. First, a fixed auxiliary array, such as a Boolean jet with at most 2^{D+2} entries, contributes a constant depending on D . Its sublevel exponent multiplies a logarithmic accuracy; it does not iterate a budget map once per entry. A degree- d array in a genuinely growing number n of variables, by contrast, contributes the power $(n+1)^d$, which is retained.

Second, the potentially long tuple loops in the rank-transfer proof propagate sharpened bounds

$$d_{\text{out}} \mid d_{\text{in}} D_0^{c_D}, \quad B_{\text{out}} \leq B_{\text{in}} \exp(P_D(n)).$$

Their logarithms are additive in the previous denominator and norm budgets. The cold costs of proving a sharpened bound are not inputs to the next sharpened output. Independent tensor children start with the same parent budget and are combined additively; new short thresholds descend in degree. Products of probabilities or moduli likewise add logarithmic costs.

The dependency chains can therefore be grouped as follows. All references in the first column are to $\mathcal{M}$.

Operations	Quantitative depth
Rational calculus and conversion; Appendices A–B	Joint linear systems and at most D layer passages; one lower-degree child in recursive returns.
Comparison, counting, and residue transfer; Appendices C–E	Degree or axis passages; fixed-degree peeling and adaptive removals paid by telescoping potentials.
Samplers and globalization; Appendices F–I	Layer, grade, and old-weight passages; closed-form sublevel bounds and fixed sampler schedules.
Rank transfer and active return; Sections 4–8	Additive tuple/cut costs, simultaneous pivots, and one lower-degree child per return.
Extraction, compression, and persistent history; Sections 9–10 and Sections 2–3 here	Macro, charge, and weight passages; joint coefficient arrays and finite meshes; descending numerical elimination.

Thus a fixed number of operation families have at most $D^{O(1)}$ successive nonlinear substitutions in their combined dependency chains. The persistent record adds no inverse invocation. Its two outer factors share one child; integer evaluation menus are meshed simultaneously; unitriangular blocks are inverted by one matrix operation. There is no iteration over the number of short values or mesh members.

For the degree-one comparison, the explicit bounded-potential argument of Appendix C has polynomial logarithmic cost; the number of peeling operations does not count as nonlinear depth. Summing errors over axes, taking products of selected denominators, and counting independent alternatives likewise preserve polynomial logarithmic budgets. Hence, after the structural path-count substitutions and elimination of all intermediate numerical records,

$$F, B \leq [H_D(I + 1)]^{D^{O(1)}} \leq \exp(D^{O(1)}).$$

The canonical-field and history construction in Proposition 2.2 uses creation-layer data only at strictly lower layers. The descending schedule of Section 10.3 of $\mathcal{M}$ therefore retains precisely the stated width dependence. $\square$

5 The density-increment calculation

We finish with the original dimension ledger and one width recurrence.

Proof of Theorem 1.1. Let F, B be as in Proposition 4.2. The original rank-cut recurrence has sufficient coordinate exponents

$$a_D = F + 1, \quad a_i = 2a_{i+1} + 1, \quad a_i = (F + 2)2^{D-i} - 1.$$

Allowing an extra factor p for cumulative cut counts, a common structural exponent is

$$a = 2^D(F + 3).$$

Thus all coordinate and cut-count inputs are $O_k(p^a)$.

Let v_i be an exponent for the width logarithm at weight i after all $T \ll_k p$ rounds. The strict higher-weight dependence allows

$$v_D = 1 + Ba, \quad v_i = 1 + Bv_{i+1} \quad (i < D).$$

Each added 1 pays the sum over rounds. Since $B \geq 2$ and $a \geq 1$,

$$v_1 \leq (2B)^D a.$$

One final polynomial evaluation and summation give total logarithmic cost and maximum logarithmic side requirement $O_k(p^A)$, where

$$A = 1 + Bv_1 \leq (2B)^{D+1}a.$$

In particular,

$$\log A \leq (D+1)\log(2B) + D\log 2 + \log(F+3) \leq D^{O(1)} = k^{O(1)}. \quad (15)$$

The fixed multiplicative density gain in $\mathcal{M}$ then gives a contradiction whenever

$$\log N \geq K_k(2 + \log(1/\alpha))^A.$$

Inverting this condition yields

$$r_k(N) \leq K'_k N \exp\left[-c'_k(\log N)^{1/A}\right].$$

By (15), enlarge one absolute constant C so that $A \leq \exp(k^C)$ for every $k \geq 3$. Decreasing the exponent to $\varepsilon_k = \exp(-k^C)$ and adjusting the k -dependent constants gives the stated bound. $\square$

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